

Warm-up

- (1) Determine the order of the differential equation $(y')^2 + 8y' + 3y = 4t$.
- (2) Verify that $2e^{-x} + x - 1$ solves the equation $y' = x - y$.
- (3) Verify that $\frac{1}{1-x}$ solves the equation $y' = y^2$.

Medium

- (1) Find all values of r such that $y(t) = t^r$ solves $t^2y'' + 13ty' + 32y = 0$.
- (2) Find a general solution to the equation $y' = xy e^x$.
- (3) Find a (particular) solution to the initial value problem $y'/x = \cos^2(y)$ if $y(0) = \pi/4$.
- (4) Find a general solution to the equation $y' = x^2 + 1 + x^2y + y$.
- (5) Find a particular solution to the equation $-2x + 4y^3\sqrt{x^2+3} \cdot y' = 0$ if $y(1) = 1$.

Stretch

(1) We used $y = ky'$ with solution $y = y_0 e^{kt}$ to model the growth of a population with initial size y_0 , but this isn't particularly realistic since it suggests the population will grow forever. If we introduce a constant K and multiply by the factor $1 - y/K$, we get what's called the *logistic equation*

$$y' = ky(1 - \frac{y}{K}).$$

What happens to the growth rate of the population when y much less than K ? When y is just a bit less than K ? Use these observations to sketch a picture of $y(t)$. (Hint: the constant K is usually called the *carrying capacity* of the population.)

- (2) Noting the logistic equation is separable, we can rearrange it and separate variables:

$$\frac{dy}{dt} = \frac{ky(K-y)}{K} \iff \frac{K}{y(K-y)} dy = k dt.$$

Use partial fractions to find the particular solution with initial value $y(0) = y_0$.